

Graph universality of the Conway–Kochen set and the norm-2 boundary for Kochen–Specker uncolorability

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We present a systematic computational survey showing that Kochen–Specker (KS) uncolorability in dimension 3 is governed by the availability of *norm-2 cancellations*: exact algebraic identities among coordinate ring elements that enable orthogonal triples. Across all quadratic fields, cyclotomic fields, cubic extensions, and finite rotation groups tested, KS sets exist only when the coordinate ring contains elements whose products participate in such cancellations. Six discrete algebraic islands are identified. Every construction achieving 31 rays—from integer, rational, mixed-field, and group-theoretic origins—produces the same orthogonality graph and the same basis hypergraph, verified by the VF2 isomorphism algorithm: the Conway–Kochen set CK-31 is a unique graph-theoretic attractor. We conjecture that CK-31 is the unique minimal realizable KS set in dimension 3.

INTRODUCTION

The smallest known Kochen–Specker (KS) set in dimension 3 is the 31-vector Conway–Kochen set [2, 4], constructed from integer coordinates. (We use the “original” KS definition [13]: every ray belongs to at least one orthogonal triple, no triple admits a $\{0, 1\}$ -coloring consistent with unit sum.) The Peres construction [3] uses $\mathbb{Z}[\sqrt{2}]$ coordinates (33 vectors), and Cabello’s recent simplest KS set [5] uses the Eisenstein integers $\mathbb{Z}[\omega]$ (33 vectors). Gould and Aravind [12] showed that the Peres and Penrose 33-vector sets—despite different coordinate fields—share the same orthogonality graph, and parametrized a continuous family with the constraint $|c|^2 = 2$. The best proven lower bound is 24 vectors [13, 14]. A natural question arises: what algebraic property determines whether a number field supports KS sets, and is the 31-vector minimum specific to the integer field or a universal barrier?

We present a systematic computational survey across number fields, showing that the availability of norm-2 cancellation identities among ring elements governs KS-uncolorability, and that the 31-vector minimum is achieved by algebraically independent constructions that all produce the same graph and hypergraph.

THE NORM-2 CANCELLATION BOUNDARY

For each number field K with ring of integers $\mathcal{O}_K$, we form a *coordinate alphabet* $\mathcal{A} = \{0, \pm 1, \pm \alpha, \pm \bar{\alpha}, \dots\}$ —a finite set of values from which ray coordinates are drawn (called “vector components” by Pavicic et al. [10]). The alphabet can often be read directly from the geometric representation of known KS sets: in the cube diagrams used to display the Conway–Kochen and Peres constructions [3, 4], the distinct coordinate values at the vertices are precisely $\mathcal{A}$. Two rays are orthogonal when $\sum_i \bar{u}_i v_i = 0$; for coordinates from a finite alphabet, this

requires an algebraic relation—a *cancellation identity*—that makes the sum vanish exactly. For the integer alphabet the key identity is $1 + 1 = 2$; for $\mathbb{Z}[\sqrt{2}]$ it is $(\sqrt{2})^2 = 2$; for the Eisenstein integers, $1 + \omega + \omega^2 = 0$. Each identity enables exact orthogonal triples (bases), and when enough bases interlock, no consistent $\{0, 1\}$ -coloring exists. (The trivial cancellation $1 + (-1) = 0$ in the minimal alphabet $\{0, \pm 1\}$ generates only 13 projective rays—far too few for a KS set; $1 + 1 = 2$ is the simplest identity that creates a large enough pool.) Our approach is to systematically construct alphabets from algebraic number fields, identify their cancellation identities, and test whether the resulting ray sets are KS-uncolorable.

We generate all rays in $\mathbb{CP}^2$ with coordinates in $\mathcal{A}$, close under Hermitian orthogonal completion (see [15] for details), and test KS-uncolorability via SAT (Glucose4 solver [8], PySAT). Minimization uses randomized greedy deletion (500 trials, seed 42) and does not guarantee minimality; reported sizes are upper bounds.

Table I summarizes results across 40+ fields. The pattern: KS sets were found only when the coordinate ring contains elements enabling *norm-2 cancellations*.

The mechanism is cancellation identities: algebraic relations among alphabet elements that make dot products vanish exactly. Across all tested fields, KS-uncolorability requires one of exactly two types of cancellation: (A) *norm-2 cancellation*—an identity producing the value 2 from products of ring elements, as in $1 + 1 = 2$ (integers), $(\sqrt{2})^2 = 2$ (Peres), and $\alpha \bar{\alpha} = 2$ (Heegner-7); or (B) *phase cancellation*—a root-of-unity sum vanishing exactly, as in $1 + \omega + \omega^2 = 0$ (Eisenstein). These are qualitatively different: norm-2 cancellations balance two small terms against one larger term, while phase cancellations sum three equal-magnitude complex terms to zero. For rings where neither mechanism is available within our bounded alphabets, we found orthogonal pairs but insufficient orthogonal triples to support KS-uncolorability. (Note: orthogonal triples do exist in these rings—e.g., $\mathbb{Z}[\sqrt{3}]$ produces 15 bases from 91 rays—but the hyper-

TABLE I. Representative results. The column $|\alpha|^2$ shows the squared modulus of the ring generator used. KS sets arise when the effective coordinate set (after completion) contains norm-2-type cancellations. “Min” is the smallest KS subset found (upper bound on true minimum).

Field / Ring	$ \alpha ^2$	KS?	Min
$\mathbb{Z}$ (integers, $\alpha = 2$)	–	Yes	31
$\mathbb{Z}[\sqrt{2}]$	2.000	Yes	33
$\mathbb{Z}[\omega]$ (Eisenstein)	1.000	Yes	33
$\mathbb{Z}[\sqrt{-2}]$	2.000	Yes	33
$\mathbb{Z}[(1+\sqrt{-7})/2]$	2.000	Yes	43
$\mathbb{Q}(\varphi)$ (golden) [†]	2.618	Yes	52
$\mathbb{Z}[\sqrt{3}]$	3.000	No	–
$\mathbb{Z}[\sqrt{5}]$	5.000	No	–
$\mathbb{Z}[\sqrt{-3}]$ ($\neq \mathbb{Z}[\omega]$)	3.000	No	–
$\mathbb{Z}[\sqrt{-5}]$	5.000	No	–
$\mathbb{Q}(\sqrt[3]{2})$ basic	1.587	No [‡]	–
$\mathbb{Q}(\sqrt[3]{2})$ extended	2.520	Yes	60

[†]Cross-product completion generates norm- ≤ 2 coordinates ($1/\varphi$ satisfies $|1/\varphi|^2 < 1$).

[‡]Basic alphabet colorable; extended alphabet $\{0, \pm 1, \pm \sqrt[3]{2}, \pm \sqrt[3]{4}\}$ is uncolorable via completion.

graph is KS-colorable.)

The golden ratio field illustrates an important subtlety: $|\varphi|^2 \approx 2.618 > 2$, but completion generates the reciprocal $1/\varphi$ with $|1/\varphi|^2 < 1$, introducing norm-2-type cancellations into the effective coordinate set. The relevant criterion is not the generator’s norm alone but whether the *generated coordinate set* (after closure) admits sufficient cancellations.

Crucially, the obstruction for $|\alpha|^2 \geq 3$ fields is *not* the absence of orthogonal triples. The rings $\mathbb{Z}[\sqrt{3}]$, $\mathbb{Z}[\sqrt{5}]$, and $\mathbb{Z}[\sqrt{-5}]$ all produce substantial pools (133 rays, 94 bases each after completion), yet remain KS-colorable. The bases exist but are not configured to force uncolorability—the cancellation structure is insufficient.

Additional evidence: among cyclotomic fields $\mathbb{Q}(e^{2\pi i/n})$ for $n \leq 30$, KS-uncolorability was observed when $6 \mid n$ —precisely the values for which the cyclotomic ring contains the Eisenstein integers $\mathbb{Z}[\omega]$ as a subring (since $\omega = e^{2\pi i/3}$ and $-1 = e^{i\pi}$ both lie in $\mathbb{Z}[e^{2\pi i/n}]$ when $6 \mid n$).

SIX ALGEBRAIC ISLANDS

Among all tested fields, six algebraic islands were found to support KS sets:

The Peres and $\mathbb{Z}[\sqrt{-2}]$ islands produce graph-isomorphic minimal sets (identical degree sequences, spectra, and WL-1 signatures), yielding five distinct graph types. The Heegner-7 and golden ratio islands are genuinely new constructions not present in any ex-

TABLE II. The six algebraic KS islands found in dimension 3. “Min” values are upper bounds from greedy deletion. Graph type indicates isomorphism class of the smallest found set.

Island	Min	Bases	Graph type
Integer (CK-31)	31	17	Unique (31v)
Peres	33	16	Norm-2 (33v)
Eisenstein	33	14	Eisenstein (33v)
$\mathbb{Z}[\sqrt{-2}]$	33	16	$\cong$ Peres
Heegner-7	43	23	New
Golden ($\mathbb{Q}(\varphi)$)	52	–	New

isting KS catalogue; details are reported in a companion paper [15].

GRAPH UNIVERSALITY OF CK-31

The central result: every algebraic or group-theoretic construction that achieves 31 rays in dimension 3 produces the *same orthogonality graph and the same basis hypergraph*.

We tested three independent constructions from algebraically distinct origins:

1. Integer alphabet $\{0, \pm 1, \pm 2\}$ (109 rays after completion)
2. Rational $\mathbb{Z}[\frac{1}{2}]$ alphabet $\{0, \pm 1, \pm 2, \pm \frac{1}{2}\}$ (109 rays)
3. Mixed norm-2: $\{0, \pm 1, \pm 2, \pm \sqrt{2}, \pm \omega, \pm \bar{\omega}\}$ (489 rays after completion)

All three resulting smallest-found sets have identical invariants:

TABLE III. Graph and hypergraph invariants of all 31-vector KS sets found. All entries are identical across all three constructions.

Invariant	Value
Vertices	31
Edges (orthogonal pairs)	71
Bases (orthogonal triples)	17
Degree sequence	$(3^4, 4^{14}, 5^8, 6^3, 8^2)$
Triangles	17
$\text{tr}(A^4)$	1250
Spectral radius	4.9443
WL-1 test	Indistinguishable

We verified pairwise graph isomorphism for all $\binom{3}{2} = 3$ pairs using the VF2 algorithm [9] (via NetworkX), which provides a definitive yes/no answer and an explicit vertex mapping. All three pairs are isomorphic. We further verified that these mappings preserve the basis hypergraph: the 17 orthonormal triples in each construction map identically under the graph isomorphism, establishing hypergraph isomorphism as well. The same orthogonality structure thus emerges from integer, rational, and

mixed-field origins. Explicit ray coordinates, edge lists, basis lists, and VF2 certificates are provided as Supplemental Material [11].

DISCUSSION

The universality of CK-31 has two implications.

Note that universality does *not* hold at 33 vectors: the Schutte–Bub set has a different orthogonality graph from the Peres/Penrose family [5, 12], so multiple non-isomorphic KS graphs coexist at that size. The fact that universality emerges at 31 is therefore non-trivial. One explanation is that CK-31 is the unique 31-vertex subgraph obtainable from the entire Gould–Aravind continuous family [12]—regardless of parameters, deleting two rays from any 33-vector member of the family yields the same 31-vertex structure.

First, this strengthens the case that 31 is the true minimum for KS sets in dimension 3: not only does no tested algebra break the barrier, but every algebra that reaches it arrives at the same graph *and* the same hypergraph. The CK-31 orthogonality structure appears to be a unique attractor across all algebraic origins.

Conjecture 1. *The Conway–Kochen set CK-31 achieves the true minimum: no KS set with 30 or fewer rays exists in dimension 3, and CK-31 is the unique 31-ray KS graph up to isomorphism.*

The proven lower bound is 24 [13, 14]; closing the gap between 24 and 31 remains the central open problem. This conjecture is supported by: the rigidity result of Trandafir and Cabello [7], who showed that every bipartite perfect quantum strategy based on 31 rays yields a unique strategy up to local isometry; our verification that 31 is the minimum found within the integer island across all alphabet sizes tested; the VF2-verified universality across three algebraically independent constructions at both graph and hypergraph level; and the failure of 30,000+ random KS-uncolorable hypergraphs to achieve geometric realizability in $\mathbb{R}^3$.

Second, the norm-2 cancellation boundary provides a *predictive heuristic*: given a number field and a bounded coordinate alphabet, one can estimate whether KS sets are likely by checking whether the ring supports norm-2-type cancellation identities among its elements. The relationship is tight: an alphabet is a viable candidate for producing a KS set only if it carries a cancellation identity that makes dot products vanish exactly among triples. Without such an identity, no exact orthogonalities arise and KS-uncolorability is impossible. The search for new KS-supporting alphabets is therefore equivalent to the search for new cancellation identities of sufficiently low norm. While this does not constitute a proof (the

search space is not exhausted), the heuristic has correctly predicted the outcome for every field tested, reducing an NP-hard combinatorial search to an algebraic check.

For two-element alphabets $\{0, \pm 1, \pm x\}$, the classification is provably *complete*: enumerating all three-term zero-sums from the product set $\{0, \pm 1, \pm x, \pm |x|^2\}$ yields exactly four non-trivial identities— $x = 2$, $|x|^2 = 2$, $|x|^2 = x + 1$, and $x^2 + x + 1 = 0$ —each corresponding to a known island [16]. Our coverage extends to quadratic fields ($\mathbb{Q}(\sqrt{d})$ for $d = 2, \dots, 30$; all nine class-1 imaginary quadratic fields) and cyclotomic fields ($n \leq 30$). The principal open question is whether multi-element alphabets from higher-degree fields harbor cancellation identities not reducible to norm-2 or phase type. The cubic field $\mathbb{Q}(\sqrt[3]{2})$ already produces a KS set at 60 vectors, suggesting that the classification may extend but with rapidly increasing cost; quartic and higher extensions remain unexplored.

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